

Keywords: agent memory • multimodal retrieval • graph neural networks • long-horizon agents

GraphMemix: Query-Aware Evidence Forests for Long-Term Multimodal Agent Memory

A Learned Forest-Construction Policy Adapts Multi-Hop Retrieval to Query Intent Without Ground-Truth Path Supervision

By Geng Li, Yuhao Wang, Dong Li, Jianye Hao, Yuxin Peng

arXiv:2608.26983 • Preprint, August 2026

Long-horizon multimodal agents accumulate heterogeneous memories — images, video clips, text segments — and must retrieve relevant evidence across modalities for each new query. Flat dense retrieval handles heterogeneity but misses multi-hop dependencies; graph memory captures dependencies but cannot adapt its retrieval structure to query intent. GraphMemix resolves this by learning a lightweight policy that dynamically constructs a query-aware evidence forest from a persistent heterogeneous memory graph, improving retrieval accuracy by up to +11.3% over dense retrieval on long-horizon benchmarks at sub-second query latency.

AT A GLANCE

Problem: Multimodal agents need memory retrieval that handles cross-modal heterogeneity and multi-hop reasoning simultaneously; neither dense retrieval nor static graph schemas do both.

Why it matters: As agents operate over longer interaction histories with diverse modalities, memory retrieval quality becomes a primary bottleneck for task completion.

Method: Heterogeneous memory graph with a GNN-based forest construction policy that selects which edges to expand, conditioned on query embedding; trained with sparse REINFORCE rewards.

Result: +7.2% on EgoSchema-Long, +11.3% on MMAgent-Bench, +6.8 F1 on MultiModal-NarrativeQA vs. dense retrieval.

Evidence: Removing query-conditioned forest construction (using static BFS) degrades by −4.3% on average; cross-modal aggregation removes another −3.1%.

The Big Picture

The memory requirements of AI agents have evolved dramatically. Early agents had short context windows and static knowledge. Contemporary multimodal agents are deployed in long-horizon settings where they accumulate interaction histories spanning hours: images from visited locations, video clips of observed events, text segments from conversations and documents.

Effective agents must retrieve relevant evidence from this growing memory on demand, for queries that are themselves multimodal and diverse in intent. A query like “where did I see the red toolbox?” requires matching a text query to visual memories. A query like “what sequence of events led to the error in step 7?” requires chaining multiple memory entries through causal links.

GraphMemix addresses this joint challenge with a single memory architecture that adapts its retrieval structure to each query.

Why Existing Approaches Fall Short

Dense retrieval: Embedding-based similarity search (DPR, FAISS) retrieves independent top- k nodes from the memory pool. This handles modality diversity (using shared embedding spaces) but treats all retrievals as independent — missing multi-hop chains where the relevant evidence at step 2 depends on what was retrieved at step 1.

Static graph memory: Systems that maintain a persistent knowledge graph (MemGraph, SAGE, GAM) capture dependencies between memory items as typed edges and support multi-hop traversal. However, their traversal strategy — usually breadth-first or heuristic depth-first — is fixed regardless of query intent. A depth-first strategy appropriate for causal chain tracing is suboptimal for gathering diverse evidence across parallel memory threads.

Hybrid systems: Retrieving from a dense index and then expanding via graph edges (retrieve-then-expand) partially addresses this, but the expansion policy is not conditioned on the query, limiting adaptation.

Core Mechanism

GraphMemix maintains a **heterogeneous memory graph** $G = (V, E)$:

- **Nodes V :** memory items typed by modality (text segment, image, video clip summary)
- **Edges E :** typed relations — temporal (A before B), co-occurrence (A and B observed together), causal (A caused B), spatial (A located near B), and co-reference (A and B refer to the same entity)

For each incoming query q , the system constructs a **query-aware evidence forest** in three steps:

Step 1 — Seed retrieval: A dual-encoder retrieves top- k seed nodes $S \subset V$ by cross-modal similarity between the query embedding h_q and node embeddings $\{h_v\}$.

Step 2 — Forest construction: A GNN-based policy π_ϕ selects which outgoing edges to expand from each seed, constructing a tree of depth d rooted at each seed. The policy decision at node v for edge (v, u) is:

$$P_\phi(u|v, q) = \sigma(W_\phi[h_v \| h_u \| h_q \| r_{vu}])$$

where r_{vu} is the relation type embedding. The top- b expansions (branch factor) are selected; the policy is applied recursively for d layers.

Step 3 — Evidence aggregation: Nodes across all trees (deduplicated) are encoded by a cross-modal transformer. A relevance scorer ranks them, and the top- m are passed to the agent.

Technical Formulation

The forest for query q from seed set S is:

$$\mathcal{F}(q, S) = \bigcup_{s \in S} T_\phi(s, q, d, b)$$

where T_ϕ is the tree built by the policy from seed s with depth d and branch factor b . Complexity is $O(|S| \cdot b^d \cdot c_\phi)$ where c_ϕ is the GNN forward pass cost — tractable for $|S| = 5, b = 3, d = 2$ (≤ 45 policy calls per query).

The policy is trained with REINFORCE using downstream task accuracy R as the reward signal and a baseline $\hat{R}$ from dense retrieval:

$$\nabla_\phi \mathcal{J} = \mathbb{E} \left[(R - \hat{R}) \sum_{e \in \mathcal{F}} \nabla_\phi \log P_\phi(e) \right]$$

No ground-truth paths through the graph are required; the policy learns which edge types to expand under which query conditions from task-level reward alone.

The cross-modal aggregation transformer uses cross-attention between text and non-text nodes at each depth level, with modality-type embeddings injected via learned positional offsets.

Learning or Inference Procedure

Training:

1. Pre-train seed encoder with contrastive cross-modal loss
2. Train forest policy with REINFORCE on each benchmark, using 80% of instances for training
3. Fine-tune cross-modal aggregation transformer on evidence re-ranking objective

Inference: Given query q , retrieve seeds (~ 10 ms), run forest construction policy (~ 50 ms for $|S| = 5, b = 3, d = 2$

on an A100), aggregate and rank evidence (~ 270 ms). Total: ~ 0.32 s per query for a 10K-node graph.

What the Guarantee Says

No formal convergence guarantees are provided for the REINFORCE training. Empirically, the policy training converges in 3–5 epochs on all three benchmarks. The system provides no worst-case latency guarantee, but median latency is 0.32s and 99th percentile is 1.1s on a 10K-node graph.

Experimental Findings

- **EgoSchema-Long:** +7.2% accuracy vs. dense retrieval; +4.1% vs. best static graph baseline
- **MMAgent-Bench:** +11.3% task completion vs. dense retrieval; +6.8% vs. static graph
- **MultiModal-NarrativeQA:** +6.8 F1 vs. dense retrieval; +3.9 F1 vs. static graph
- **Latency:** 0.32s median; scales sub-linearly with graph size to 100K nodes

Ablations and Interpretation

Query-conditioned vs. static forest construction: Replacing the policy with static BFS expansion degrades by -4.3% on average — confirming that query-aware adaptation of tree structure is the primary driver of improvement.

Cross-modal aggregation: Replacing the cross-modal transformer with modality-specific encoders (no cross-attention) degrades by -3.1% , confirming that cross-modal evidence fusion adds value beyond independent retrieval.

Depth d and branch factor b : $d=2, b=3$ is optimal; $d=3$ slightly improves multi-hop tasks but increases latency by $3\times$. $b=1$ (DFS) is best for causal-chain tasks; $b=5$ (broad BFS) is best for evidence-gathering tasks; the policy learns to approximate this per-query adaptively.

Reward baseline: Using dense retrieval performance as REINFORCE baseline substantially stabilises policy training compared to a constant baseline (variance reduced by $4.1\times$).

Reference

Geng Li, Yuhao Wang, Dong Li, Jianye Hao, Yuxin Peng. **GraphMemix: Query-Aware Evidence Forests for Long-Term Multimodal Agent Memory.** arXiv:2608.26983, August 2026. <https://arxiv.org/abs/2608.26983>